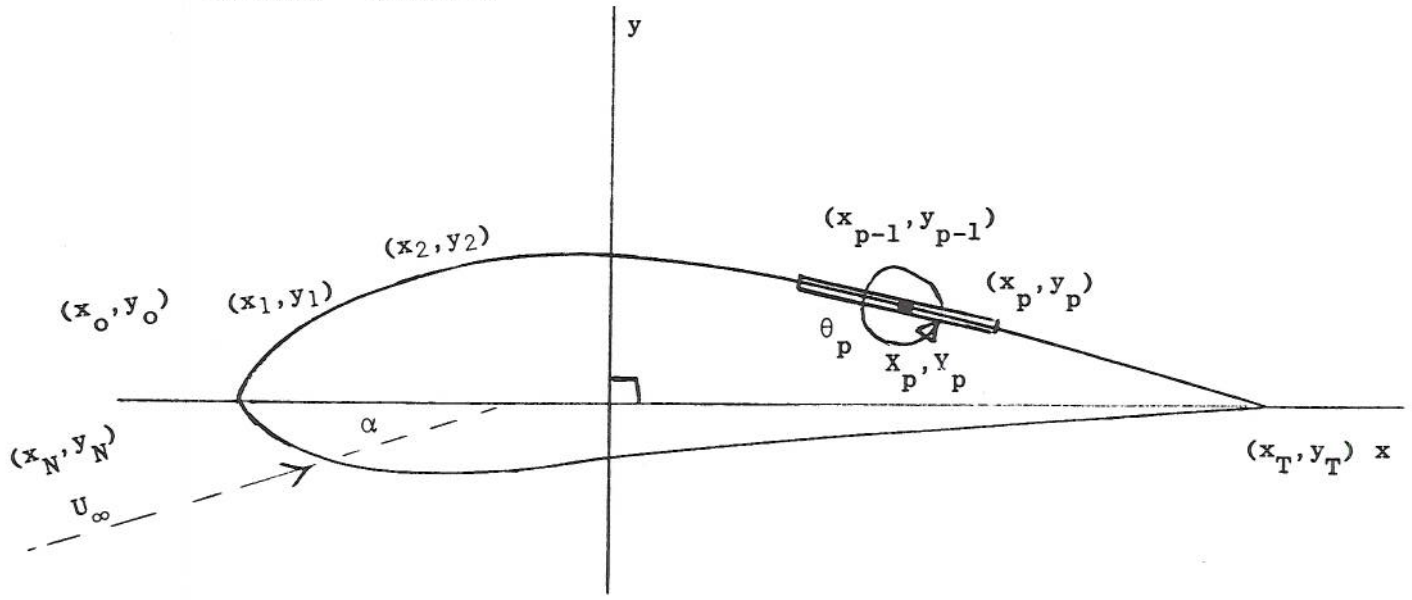
DEPARTMENT OF AERONAUTICS

PART III AERODYNAMICS III

Further Notes on Surface Singularity Methods

The body co-ordinate system - finding g(p,q)

Example - Aerofoil



Chord line is x axis; y axis can be any line perpendicular to this. Aerofoil is divided up into points in clockwise cyclic order (x_o, y_o) $(x_1 y_1)$ - - - (x_N, y_N) where x_N, y_N coincides with (x_o, y_o).

Choose (x_o, y_o) anywhere (leading edge is usual) but if the body is an aerofoil <u>one</u> of the points must be the trailing edge.

The pth element joins (x_{p-1}, y_{p-1}) to (x_p, y_p).

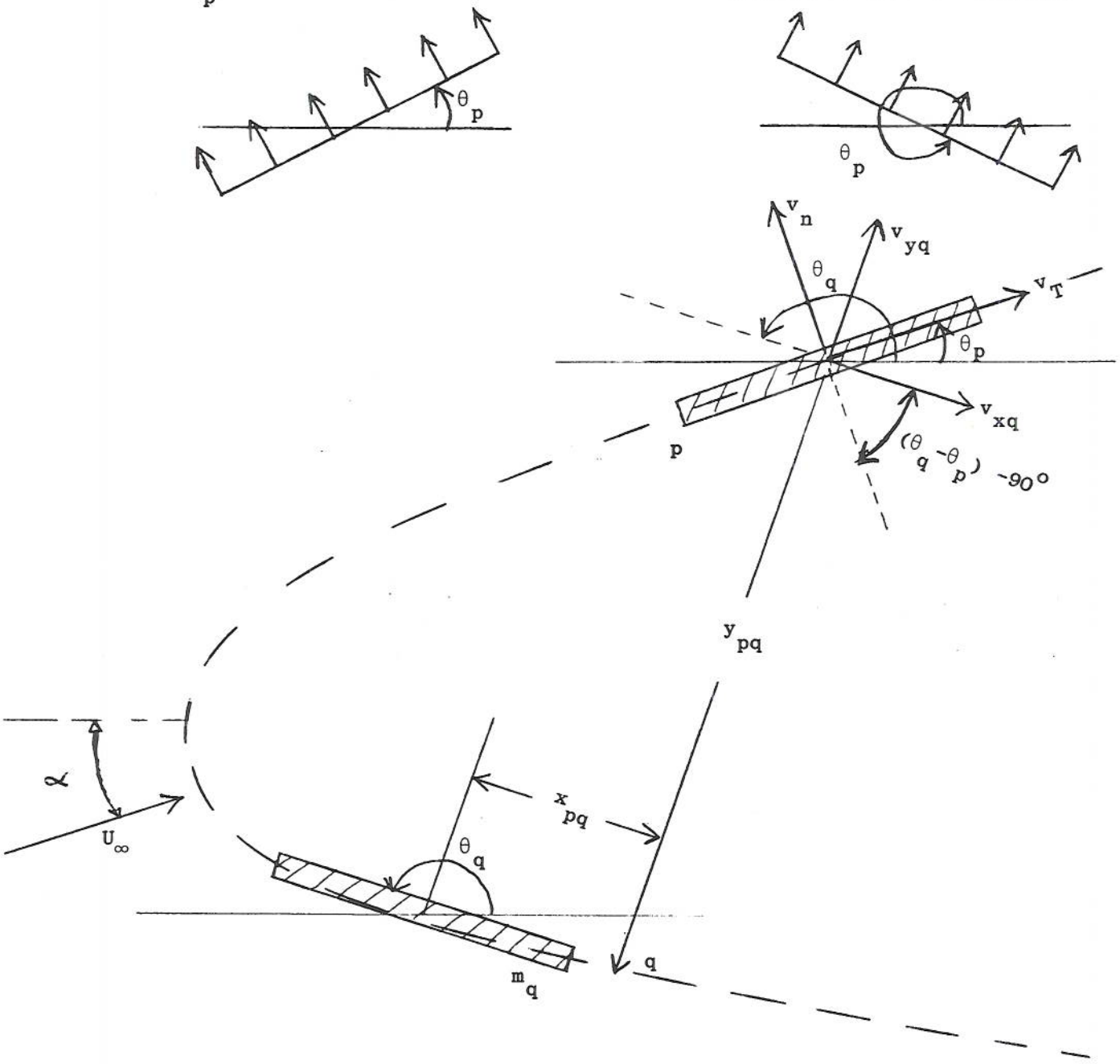
It has source density m_p.

Length $\Delta s_p = \left[\left(x_p - x_{p-1}\right)^2 + \left(y_p - y_{p-1}\right)^2\right]^{\frac{1}{2}}$.

Control point $X_p, Y_p = \dfrac{x_p + x_{p-1}}{2}, \dfrac{y_p + y_{p-1}}{2}$.

Angle of element to x axis θ_p.

θ_p measured anticlockwise from x axis to <u>inside</u> surface of element.



Find velocity components at the control point of the pth element due to the qth element.

normal velocity $v_n(p,q) = - v_{xq}\sin(\theta_q-\theta_p) + v_{yq}\cos(\theta_q-\theta_p)$

tangential velocity $v_t(p,q) = v_{xq}\cos(\theta_q-\theta_p) + v_{yq}\sin(\theta_q-\theta_p)$ where v_{xq} and v_{yq} can be found from equations (1) and (2)

$$v_{xq} = \frac{m_q}{4\pi} \ln \left(\frac{\left(x_{pq} + \frac{\Delta s_q}{2}\right)^2 + y_{pq}^2}{\left(x_{pq} - \frac{\Delta s_q}{2}\right)^2 + y_{pq}^2} \right)$$

and

$$v_{yq} = \frac{m_q}{2\pi} \tan^{-1} \left(\frac{y_{pq} \, \Delta s_q}{x_{pq}^2 + y_{pq}^2 - \left(\frac{\Delta s_q}{2}\right)^2} \right)$$

Hence combining the above we can write

$$v_n(p,q) = Ap,q \quad m_q$$

$$v_t(p,q) = Bp,q \quad m_q$$

Ap,q and Bp,q are known as influence coefficients and will have to be computed for every combination of p and q.

The velocity components at p due to the free stream U_∞ at angle α are

$U_n = U_\infty \sin(\alpha - \theta_p)$ and $U_t = U_\infty \cos(\alpha - \theta_p)$.

The equation for each control point is now

$$\sum_{q=1}^{N} Ap,q \; m_q = -U_\infty \sin(\alpha - \theta_p)$$

The values of m can be found by standard matrix inversion techniques

$$m_q = -\sum_{p=1}^{N} \left[Apq\right]^{-1} U_\infty \sin(\alpha - \theta_p)$$

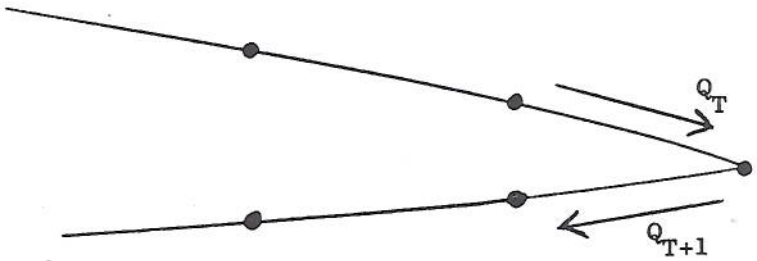
where $\left[Apq\right]^{-1}$ is the inverse matrix of A_{pq}.

Now the velocities tangential to the surface Q can be calculated since

$$Q_p = \sum_{q=1}^{N} Bp,q\ m_q + U_\infty \cos\ (\alpha - \theta_p)$$

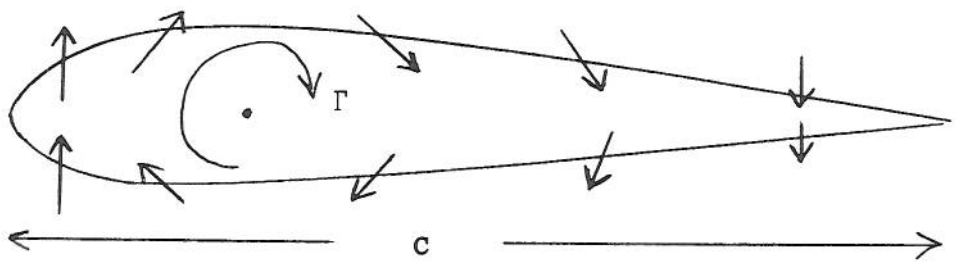
and also the pressure coefficient where $C_{p_p} = 1 - \frac{Q^2{}_p}{U_\infty{}^2}$. This completes the solution for an arbitrarily shaped 2D <u>non-lifting</u> body.

No vortices have been used and hence no circulation (and lift) have been generated. The Kutta condition has not been satisfied and in general there will be a velocity difference on the elements either side of the trailing edge.



i.e. $Q_T + Q_{T+1} \neq 0. = \Delta Q_{t.e}$

However in potential flow we can superpose solutions and we can add sufficient circulation to make $\Delta Q_{te} = 0$. Let us now consider solving for the source distribution required on an aerofoil when the onset flow is no longer the free stream but a unit strength vortex placed somewhere inside the aerofoil.



let $\frac{\Gamma}{U_\infty c} = 1.$

The problem to be solved is

$$\sum_{q=1}^{N} Ap,q\ m_{\Gamma q} = 1\ U_{np\Gamma}$$

where $U_{np\Gamma}$ is the normal velocity on element p due to the unit strength circulation. This leads to another set of sources m_Γ which make the aerofoil a streamline when unit circulation is placed around it, $m_{\Gamma q} = -\sum_{p=1}^{N} [Apq]^{-1}\ U_{np\Gamma}$. The tangential surface velocity Q_Γ due to the sources m_Γ and the circulation is given by $Q_{p\Gamma} = \sum_{q=1}^{N} Bp,q\ m_{q\Gamma} + U_{tp\Gamma}$, $U_{tp\Gamma}$ is tangential velocity due to vortex.

This distribution will not satisfy the Kutta condition either and there will be a velocity difference across the trailing edge $Q_{\Gamma te} = Q_{T,\Gamma} + Q_{T+1,\Gamma}$. However we can multiply the circulation by a factor K, (and hence all the sources $m_{q\Gamma}$ will increase by a factor K) and add the solution to the solution for the original source distribution m_q and still keep the surface a streamline. We need to choose the constant K such that

$$K\Delta Q_{\Gamma t.e} + \Delta Q_{t.e} = 0 \text{ at the trailing edge.}$$

The Kutta condition will now be satisfied and the surface velocity at the pth element will be

$$\hat{Q}_p = Q_p + KQ_{p\Gamma}$$

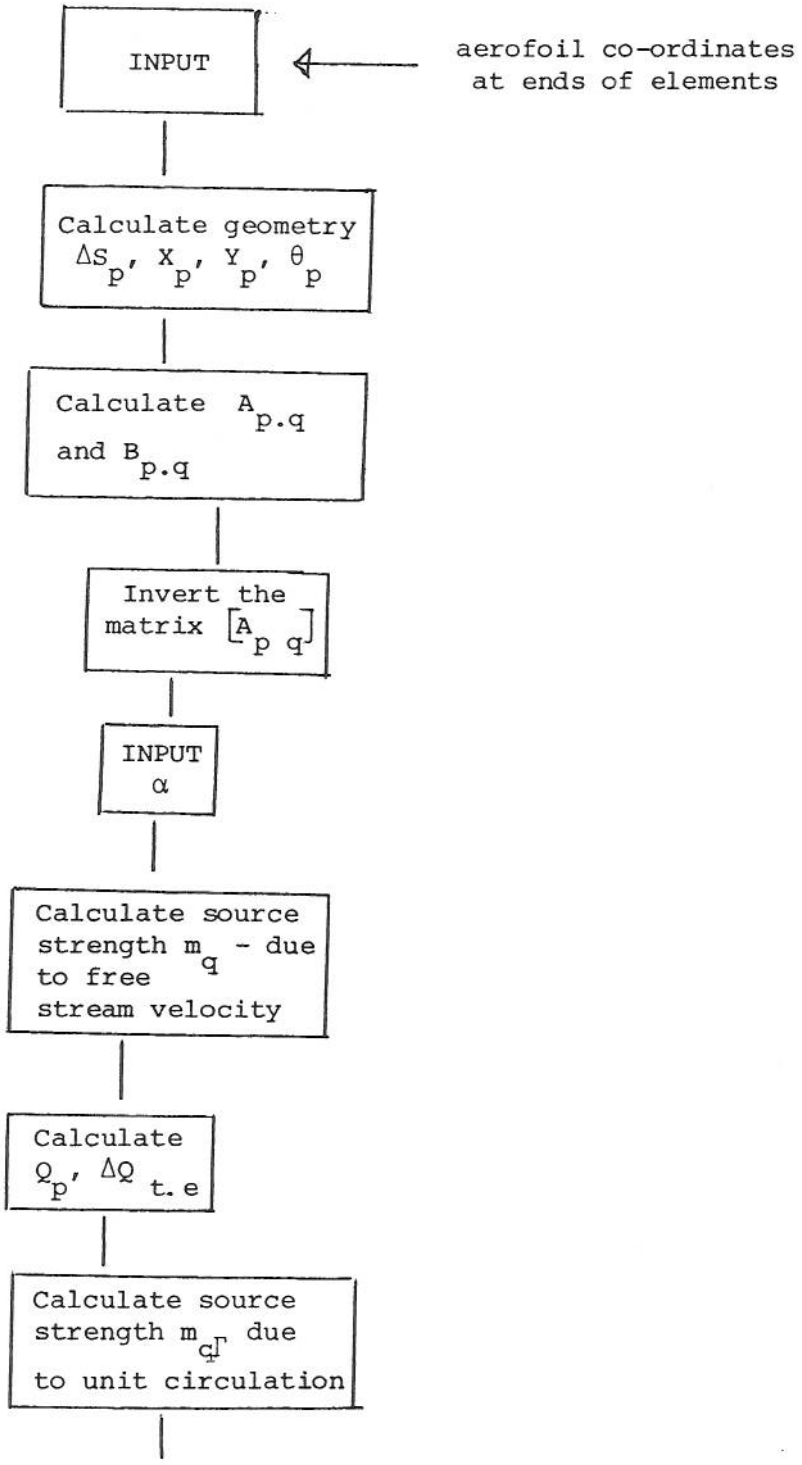
$$\hat{Q}_p = Q_p - \frac{\Delta Q_{t.e}}{\Delta Q_{\Gamma t.e.}} \cdot Q_{p\Gamma}$$

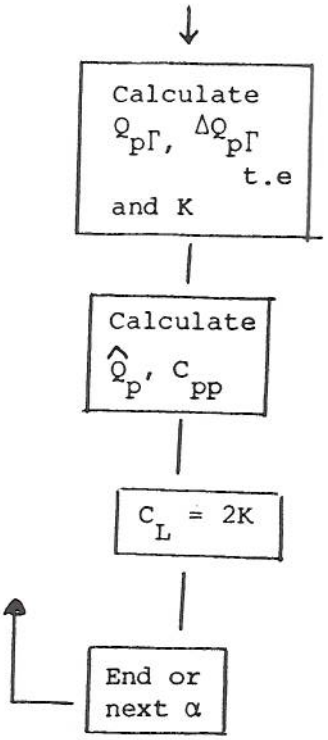
$$\text{Lift} = \rho U_\infty \Gamma \quad , \ \Gamma = KU_\infty c$$

$$\therefore \ \text{Lift} = \rho U_\infty{}^2 Kc$$

$$C_L = 2K$$

Flow chart for a computer program to calculate 2D aerofoil flow





Notes

(1) A large variety of linear boundary value problems can be computed in this way, a single body, multiple bodies - slotted aerofoils, aerofoils near the ground, hydrofoils beneath a water surface. The method can be used to solve the wave equation for example for wave diffraction around off-shore structures. It is applicable where viscous effects are either unimportant or can be dealt with in a simpler way such as by the addition of a circulation.

(2) Three-dimensional bodies can be treated by using surface source elements (i.e. using square or rectangular panels) so

long as the body is non-lifting. Lifting problems in 3D are more complex since a trailing vortex wake is shed and very lengthy iteration processes are needed to obtain its correct position.

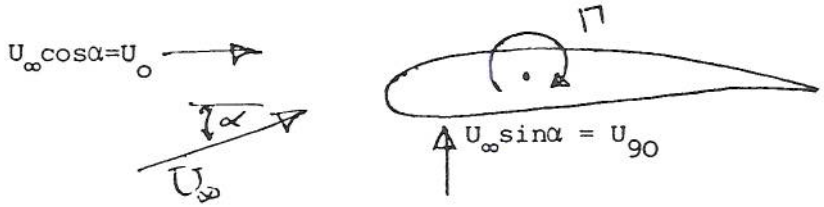
(3) For 2D aerofoils the single vortex inside the aerofoil is often replaced by unit vorticity distributed along each of the same elements, as the sources. This has the advantage that the influence

functions for the vorticity distribution are related to those for the source distribution

$$A_{p,q\Gamma} = B_{p,q} \text{ (sources)}$$

$$B_{p,q\Gamma} = -A_{p,q} \text{ (sources)}$$

(4) Since we can superpose solutions in potential flow the flow around an aerofoil at any incidence α can be made up of a flow at zero incidence, a flow at 90° incidence and a circulation



The general solution for α° will be

$$\hat{Q}_{p\alpha} = Q_{po^\circ}\cos\alpha + Q_{p90^\circ}\sin\alpha - \frac{\Delta Q_{t-eo^\circ}\cos\alpha + \Delta Q_{t-e90^\circ}\sin\alpha}{\Delta Q_{\Gamma}\ t.e}\, Q_{p\Gamma}$$

where Q_{po° is surface velocity at p due to unit free stream at $\alpha = 0^\circ$

Q_{p90° " " " " " " $\alpha = 90^\circ$

$Q_{p\Gamma}$ " " " unit circulation

(5) The lift can be obtained directly from the circulation thus

$$L = -\frac{\rho U_\infty^{\ 2}}{\Delta Q_{\Gamma t\ e}} \left(\Delta Q_{t.eo^o} \cos\alpha + \Delta Q_{t.e90^o} \sin\alpha\right) \quad C$$

(6) The more elements around the aerofoil the more accurate the solution. Ideally the points at the ends of the elements should be spaced such that if the aerofoil were transformed into a circle the points would be evenly spaced around the circle. In practice we have more points near regions of high curvature i.e. near the leading and trailing edges. Remember that we dont calculate pressures at these points but at the mid points of the elements. A.M.O. Smith suggests non-lifting 2D bodies and axisymmetric bodies 60-80 elements, 2D aerofoils 100, 3D bodies 500 → 1000 panels.

(7) We have used straight line elements with uniform distributions of singularities however varying strength can be used together with curved elements. Generally this leads to more algebra but less elements are required and computing time is reduced. Not worth it for 2D but could be important for 3D bodies.